

IMPERIAL

Supply Chain Game Coursework

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A. Introduction

This report presents the inventory strategy developed for a supply chain simulation, operating within a fictitious continent. The network was structured around two key elements, namely, warehouse operation and factory expansion. The primary objective was to maximise profit by effectively satisfying demand within a starting budget constraint.

The first section of this document covers the single region run, focusing solely on Calopeia, where decisions centred around the capacity, reorder point, order quantity, and shipment method. The second addresses the full multi-region network, which spanned five regions: Calopeia, Sorange, Entworpe, and Tyran, alongside the small island of Fardo, where demand patterns varied across regions and decisions extended to the optimal placement of factories and warehouses and more complex interactions. In both runs, the analysis is structured sequentially: long-term strategic decisions, such as network configuration and capacity, are addressed first; these are followed by short-term operational policies governing daily inventory and shipment decisions. The third section presents a conclusion drawing on the key learnings from both runs and evaluating overall performance.

The team's decision-making process was grounded in the analysis of two years of historic demand data provided for each run. Collective brainstorming sessions were used to interpret demand patterns, align on priorities, and agree on a shared strategic direction. Python was employed to support these data-driven decisions through simulation. During both runs, short-term operational decisions were implemented automatically using a bot, which was programmed with predefined decision rules for each run and executed a decision cycle once per in-game day to update the simulation's parameters as required.

B. Single Region Run

B.1. Demand Analysis

Daily demand for the previous two years ($n = 730$) was first analysed to establish its key characteristics and guide the selection of an appropriate forecasting method. Table 1 presents a statistical summary of historic demand, describing its central tendency and dispersion.

Table 1. Historic Demand Statistic Summary	
Statistic	Historic data
Mean	39.19
Median	36.50
Standard deviation	26.49

To assess whether demand could be reasonably treated as normally distributed, visual normality checks were conducted using a histogram and a Q–Q plot (see Appendix A). The histogram indicated a pronounced right-skew, while the Q–Q plot showed systematic deviation from the normal reference line, particularly in the upper quantiles. These observations suggested that demand was not well represented by a normal distribution, and therefore a forecasting method was required that could accommodate non-normal and structured patterns in the data.

Following this, the time series was inspected for temporal structure. Visual examination across the 730-day period indicated a repeating yearly pattern, suggesting strong seasonality. To distinguish predictable seasonal variation from random noise, the coefficient of variation (CV) was evaluated for both the raw series and the deseasonalised residuals obtained via Seasonal-Trend decomposition (STL). The naïve series exhibited a relatively high CV of 0.676, indicating substantial overall variability. After deseasonalisation, the CV decreased markedly to 0.177, confirming that most of the demand variability was systematic and seasonal rather than purely random.

Given the presence of strong and stable seasonality and limited evidence of a sustained trend, Holt–Winters additive exponential smoothing was selected as the forecasting approach, as given by Equation 1:

$$\hat{y}_{t+h} = L_t + h \cdot T_t + S_{t-m+h} \tag{1}$$

where L_t , T_t , and S_t represent the level, trend, and seasonal components, updated recursively using smoothing parameters α , β , and γ . The fitted parameters and model performance metrics are reported in table 2.

Using the estimated parameters, the forecasted demand was projected over the simulation horizon, as illustrated in Figure 1. Despite the smoothing parameters being zero, it was decided to keep using this forecast, as Holt–Winters provided enough flexibility to allow the system to overcome any uncertainty in demand and recover. The model forecast shows that the seasonal pattern continues beyond day 730 into the simulation period. Two clear demand peaks were also

identified in the upcoming two years. This allowed the team to anticipate higher demand periods in advance and prepare inventory and capacity decisions accordingly to ensure those peaks could be satisfied.

Table 2. Holt-Winters fitted parameters and model fit statistics (730-day history)

Parameter	Symbol	Value	Interpretation
Level smoothing	α	0.0304	Level evolves slowly: High weight on prior estimate
Trend smoothing	β	0.0000	No significant trend in demand level
Seasonal smoothing	γ	0.0000	Seasonal pattern is stable across both years
Initial level	L_0	40.90	Baseline demand (dr/day)
Initial trend	T_0	-0.0124	Negligible downward drift

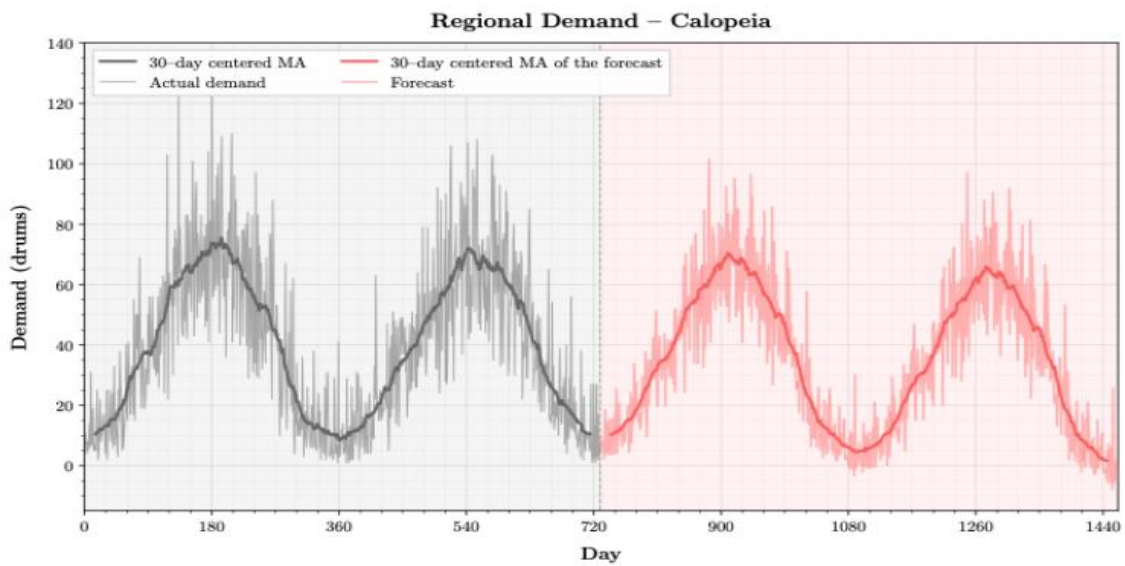


Figure 1. Four-year Demand Cycle

B.2. Strategy Development

B.2.1. Long-term Decisions

Following the demand forecast, the next step was to determine the appropriate factory capacity required to satisfy the anticipated demand peaks while avoiding unnecessary capital expenditure.

A key trade-off emerged between expanding capacity and managing inventory. Factory capacity expansion costs were substantial at \$50,000 per drum per day (\$/dr/day), whereas finished goods holding costs were relatively negligible at \$90 per drum per year (\$/dr/year). This cost structure implied that underestimating demand and losing sales would be significantly more expensive than holding excess inventory. Consequently, the strategy prioritized maintaining high factory utilization, particularly during low-demand periods, in order to build inventory ahead of peak seasons.

Financial Rationale for Inventory Build-Up

The decision to build inventory during low-demand periods was supported by cost asymmetry within the simulation environment. The holding cost was calculated using Equation 2, while the profit per drum was computed through Equation 3.

$$h = \frac{90}{365} \approx \$0.25/\text{dr}/\text{day} \quad (2)$$

$$\pi = 1450\$ - \frac{2000\$}{200} - 900\$ - \frac{15000\$}{200} - 150\$ = \$315/\text{dr} \quad (3)$$

Given this margin, a drum could be held for $\frac{315}{0.25} = 1,260$ days before holding costs would exceed the profit missed from a lost sale. Therefore, maintaining continuous production and accumulating inventory ahead of forecasted peaks was economically justified.

Strategy Simulation

A Python-based bot was developed to replicate the behaviour of the supply chain simulation. Demand from day 730 onward was simulated using the same statistical characteristics mean of 39.19 drum per day (dr/day) and CV of 0.676 as the historical two-year dataset, ensuring consistency with the observed demand structure.

Using this tool, several capacity levels were tested and assessed based on total cash balance, cumulative lost sales, inventory stability, and demand fulfilment performance. The results indicated that a factory capacity of 50 dr/day produced the most favourable outcome by achieving the highest final cash balance, while minimising lost sales. Lower capacity levels led to unmet demand during peak periods, whereas higher levels increased capital costs without delivering a significant financial improvement.

Capacity expansion would therefore be implemented immediately at day 730, at the start of the simulation horizon. This ensured that inventory accumulation could begin early in the forecast cycle, maximising capital utilisation and reducing the risk of stockout during the first anticipated demand peak. Delaying expansion would have limited the ability to build inventory in advance and increased the likelihood of lost sales.

B.2.2. Short-term Decisions

Daily operational decisions were executed through the automated bot (see Appendix B for decision-making flowchart). Each in-game day, it executed a structured decision cycle: It first scraped the current system state, generated an updated demand forecast, selected the appropriate operational mode, computed the required inventory parameters, and finally applied the updated settings within the simulation. Based on this process, the reorder point (ROP) and order quantity (Q^*) were recalculated using the latest demand forecast, rolling demand variability, and effective lead time.

Safety Stock

The first variable the bot had to compute was the required safety stock, based on the expected demand patterns for the next 30 days and lead times (Equation 4).

$$SS = Z_{0.999} \times \sigma_d \times \sqrt{L_{eff}} \times 2 \quad (4)$$

where $Z_{0.999} \approx 3.09$, σ_d is the rolling 30-day demand standard deviation, and $L_{eff} = \frac{Q}{C} + T_{ship}$ is the effective lead time (production time + shipping time¹). The “×2” multiplier is justified by the cost asymmetry.

Future Deficit

The bot then estimated the future production deficit to determine the inventory that had to be accumulated to meet demand during the periods in which it would exceed capacity through Equation 5. A six-month (182-day) lookahead horizon was used to capture the next seasonal peak, rather than relying solely on the point forecast. A 30% buffer was calibrated after running multiple simulations for different safety adjustments, to account for forecast uncertainty and demand volatility. This buffer increased resilience during peak periods and reduced both the frequency and magnitude of stockouts. The objective was to minimise lost sales while maintaining economically efficient inventory levels.

$$D_{fut} = 1.3 \times \sum_{i=1}^{182} \max(0, \hat{y}_{t+1} - C) \quad (5)$$

Re-order Point

Using the previously calculated vales, the bot automatically selected one of three operational modes each day (see table 3), depending on mean forecasted demand over the next 14 days ($\bar{d}_{14}$), as well as the target inventory level (I^*), which was defined by the following identity:

$$I^* = D_{fut} + SS$$

Based on the mode, the bot would compute the ROP using the corresponding equation:

$$ROP \begin{cases} \sum_{i=1}^{L_{eff}} \hat{y}_{t+1} + SS & Chase \\ \max(D_{fut}, \sum_{i=1}^{L_{eff}} \hat{y}_{t+1} + SS) & Build \\ 3000 & Drawdown \end{cases} \quad (6)$$

Note that the ROP for drawdown mode is much higher than any achievable inventory. This was verified via simulation and done purposefully to ensure that the factory would be constantly producing.

¹ Shipping time = 7 days truck, 1 day mail

Table 3. Bot operating mode selection logic

Mode	Condition	Behaviour
Chase	Otherwise	Steady state; production $\approx$ demand
Build	$\bar{d}_{14} \leq C, \quad D_{fut} > 0, \quad I_t < I^*$	Accumulate stock for upcoming peaks
Drawdown	$\bar{d}_{14} > C$	Factory at 100%; deplete inventory to cover excess demand

Order Quantity and Shipping

Order quantity decisions were constrained by transportation efficiency. As trucks operate in fixed loads of 200, the batch size was set to $Q = 200$ to ensure full truck utilisation and avoid partially filled shipments. This minimised unit transport costs while keeping the factory operationally agile. Larger batch sizes (e.g. 400 or 600 units) were avoided to ensure a steady flow of drums from the factory to the warehouse. The profit per drum was calculated using Equations 7 and 8, for truck and mail shipment respectively.

$$Profit_{Truck} = \$1,450 - \frac{2000\$}{200} - \$900 - \$150 - \frac{\$15,000}{200} = \$315/dr \quad (7)$$

$$Profit_{Mail} = \$1,450 - \frac{2000\$}{200} - \$900 - \$150 - \$150 = \$240/dr \quad (8)$$

As forecasted demand declined linearly toward zero, all shipments were made via mail. Variable shipment quantities were calculated to reduce inventory to zero by day 1457, aligning with the expected demand exhaustion by day 1460. This prevented excess terminal inventory while maintaining fulfilment performance.

C. Multi-Region Network Run

The multi-region run expanded the supply chain from a single region to a full five-region network, spanning a continent (Calopeia, Sorange, Tyran, Entworpe) and an island (Fardo). In addition to the day-to-day ordering logic, this network also required investment decisions that were not present in the single region run, namely, where to place factories and warehouses, as well as how to configure facility supply and demand fulfilment between regions.

C.1. Demand Analysis

As in the first run, the first step was to analyse the demand patterns for each region. The observed behaviour demonstrated clear heterogeneity across the network, precluding the use of a single forecasting model. A region-specific analysis was therefore conducted to understand the underlying characteristics of the demand in each market.

Given Calopeia's strong seasonality and that specific values exactly matched the single region run, it was decided to use the data collected during the second run, enabling a perfect forecast. Sorange presented a challenge with only 60 days of history and a CV close to 1.0; literature on the same scenario (Merkle et al, 2022) confirmed a linear growth pattern. Based on these observations, a linear regression model was developed, as shown in Table 4. Tyran and Fardo on the other hand, showed a stable demand pattern with no significant seasonality or trend, so forecasting was done using a simple moving average from day 670 onwards. For Entworpe, the high coefficient of variation justified the use of a fixed daily rate, enabling inventory build-up when demand was non-existent to cover peak periods when they occurred. Table 4 summarizes the key statistical properties of each region, while Figure 2 shows the forecasted demand for all regions.

Table 4. Demand indicators and forecasting method for network run

Region	Pattern	Avg. Demand (dr/day)	Coefficient of Variation	Forecast method
Calopeia	Seasonal	~39	0.177	Known demand file (full 4-year series from Run 2)
Sorange	Linear growth	~75	0.996	Linear formula: $\hat{d} \approx 0.165(t - 730) + 14.2$
Tyran	Stable	~20	0.879	Moving average from day 670 onward
Entworpe	Lumpy batch (0 or 250 drums)	~15	4.43	Fixed daily rate estimate
Fardo	Stable	~16	1.068	Moving average from day 670 onward

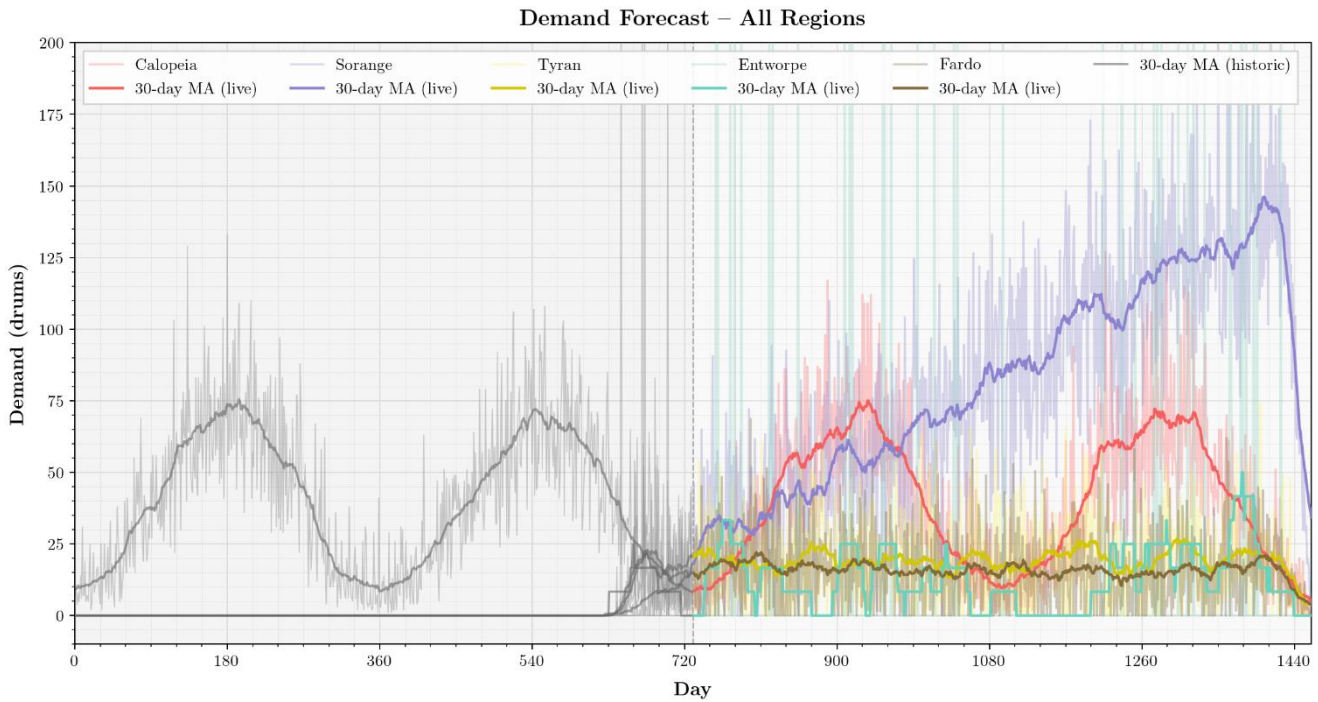


Figure 2. Demand forecast for all regions.

A final consideration in the demand analysis was how to account for the drop in demand to zero that would occur by day 1460. To factor this, all region forecasts applied an end-game multiplier in the final 30 days of the simulation (days 1430 –1460) to smoothly wind down production using Equation 9.

$$\hat{d}_r^{\text{adj}}(t) = \hat{d}_r(t) \cdot \frac{1460 - t}{30} \text{ for } t \in (1430, 1460] \quad (9)$$

C.2. Strategy Development

C.2.1. Long-term Decisions

Based on the previous demand analysis, the overarching principles that would guide the supply chain strategy were defined. Given the observed demand patterns, production was structured around maintaining a constant pipeline of drums, as explained further on. Second, an analysis was conducted to quantify how profit per drum varied across alternative transport and fulfilment strategies. Finally, the network configuration was decided based on infrastructure return-on-investment (ROI) and lead times (90 days for factories and 60 for warehouses), as well optimal production allocation configurations.

“Always On” Production Strategy

The *Always On* production strategy i.e. running every factory at 100% capacity until a shutdown trigger, was adopted as the overarching approach. Unlike the single region run’s Build/Chase/Drawdown framework, which required accurate medium-term forecasts, the multi-region run used a simpler structure to avoid frequent mode switching, minimize order and transports costs, and take advantage of the low holding costs (\$0.25/dr/day). This also fostered the network’s resilience and ensured maximum capital utilisation. Table 5 summarizes the advantages of this strategy.

Table 5. Rationale for the Always on Strategy

Advantage	Detail
Uncertainty buffer	High regional stochasticity makes mode-switching unreliable. Forecast errors cause premature mode changes and stockouts.
Fulfilment safety net	If the nearest warehouse stocks out, a more distant warehouse covers the order. Large inventories protect fill rate network wide.
Capital utilisation	Producing at 100% for ~95% of the game maximises return on factory/warehouse investment.
Deferred shutdown	The only mode transition (to shutdown) is delayed until cumulative inventory covers remaining demand, when short-horizon forecasts are most reliable.

Profit Variability

To determine the impact that different network configurations might have and get guiding principles on infrastructure construction, it was necessary to see how the different transport and fulfilment costs would affect profits. To this end, the profit per drum was calculated under different supply and fulfilment configurations (assuming order size of 200 and truck shipping), as shown in table 6. It is important to note that even the lowest-margin non-intercontinental route can hold a drum for over 3 years before incurring a loss, reinforcing the importance of the *Always On* strategy. It was also noted that servicing the demand at Fardo from the mainland reduced drum profitability by more than 70%, even when compared with cross-region warehouse fulfilment.

Table 6. Drum profitability under different shipping configurations.

Scenario	Profit	Hold Break-even
Same-region factory and warehouse	$\pi = 1,450 - \frac{2,000}{200} - 900 - \frac{15,000}{200} - 150 = \$315/dr$	1,260 days
Cross-region factory to warehouse	$\pi = 1,450 - \frac{2,000}{200} - 900 - \frac{20,000}{200} - 150 = \$290/dr$	1,160 days
Cross-region warehouse fulfilment	$\pi = 1,450 - \frac{2,000}{200} - 900 - \frac{15,000}{200} - 200 = \$265/dr$	1,060 days
Mainland factory to island warehouse	$\pi = 1,450 - \frac{2,000}{200} - 900 - \frac{45,000}{200} - 150 = \$75/dr$	300 days

Network Configuration

Based on the cost asymmetry of drum transport and order fulfilment, the next step was to determine the network configuration that would minimize stockouts and capital expenditure.

An ROI analysis was conducted to determine whether it was profitable to invest in infrastructure in a given region, by adapting the single-region run simulator so it could capture different network configurations. This enabled the team to determine which infrastructure layout was optimal, based on the previously made demand forecast for the period comprised between day 820 (when new factories and capacities became operational) and day 1460. For the mainland, the question of production allocation presented two alternatives: Capacity could be centralised in the existing Calopeia factory through a significant expansion, with finished goods distributed to regional warehouses at higher transport cost; alternatively, new factories could be built to service demand in a specific region, which would lead to savings in transport cost but would also require a significant investment (\$500,000 per factory plus an additional \$50,000 per unit of capacity i.e. dr/day). Furthermore, the decision had to be made on which regions to build a warehouse, depending on whether the reduction in fulfilment costs was greater than the capital expense of \$100,000 per warehouse and the corresponding opportunity cost from lost interest.

In the case of Fardo, the key consideration was whether the investment required to build a factory and warehouse would generate a positive ROI given its demand levels. Intercontinental shipment costs and transport times meant that supplying from the mainland should be avoided as much as possible, given the opportunity cost of not selling that same drum in an adjacent region. Although ROI was positive in all cases, see table 7, the relatively low return associated with a warehouse in Entworpe made it an unattractive investment, particularly when accounting for the demand stochasticity and lost interest. On the other hand, table 8 presents the final infrastructure configuration implemented in each region.

Based on the findings of the ROI analysis and the final infrastructure decisions in each region, the next consideration was how to configure the distribution network. This process involved three main stages for the mainland – Fardo was always treated as an independent system; however, investment there was postponed until day 755 to ensure sufficient cash was available to finance production during the initial operating days.

The first challenge was how to meet initial mainland demand before the new factories and additional capacity became operational. To address this problem, it was decided to use the starting inventory and high capacity at Calopeia to serve Tyran, Entworpe and Sorange as a unified system. This was feasible because the combined average daily mainland demand for the period between days 730 and 820 was 74 dr/day, comparable to the initial production capacity of 70 dr/day, supported by an initial inventory of 2,448 drums.

After day 820, once the factory at Sorange became operational, the long-term allocation shown in table 9 was implemented. Calopeia continued serving Tyran and Entworpe, while Sorange, similar to Fardo, operated as an isolated system. This policy allowed inventory to accumulate in Sorange during the first year, when demand remained below production capacity; the accumulated stock could then be used to meet excess demand in later stages of the simulation.

Table 7. Infrastructure investment decisions and ROI analysis.

Region	Decision	Capital Expenditure	Key Rationale	Estimated ROI
Calopeia	• Expand factory capacity	-	The existing capacity means it can easily be transformed into a production centre that can meet the demand of several regions.	-
Sorange	• Build a factory • Build a warehouse	• \$500k* • \$100k	Using local shipment and fulfilment will lead to significant savings as opposed to cross-region policies. They are expected to be approximately \$1M. It is the region with the highest expected demand (150 dr/day).	~\$424k
Tyran	• Build a warehouse	• \$100k	Demand (20 dr/day) is high enough that the savings derived from local fulfilment could justify a warehouse, but not the construction of a factory.	~\$101k
Entworpe	• No additional infrastructure	• \$0	The erratic demand patterns means that a dedicated warehouse would require a very high level of safety stock, so it is more convenient to serve it from the warehouse in Calopeia using cross-fulfilment.	\$20.6k Therefore, warehouse construction is not recommended.
Fardo	• Build a factory • Build a warehouse	• \$1,500k • \$100k	The high costs of intercontinental shipping (\$45,000/truck) means that profits are decreased to an unacceptable level (\$75/dr), so it is better to build a dedicated factory and warehouse and rely on local transport and fulfilment.	~\$1.16M

*Capacity expansion costs are not included as that money would have otherwise been invested in the factory at Calopeia.

Table 8. Warehouse-to-region demand allocation and warehouse supply policy

Region	Factory	Warehouse	System	Supplier	Notes
Calopeia	Yes (75 dr/day)	Yes	A	Calopeia	Entworpe served via cross-fulfilment no dedicated Entworpe WH built.
Tyran	No	Yes	A	Calopeia	Local fulfilment saves \$50/dr vs. cross-regional; sustained by Calopeia factory.
Entworpe	No	No	A	Calopeia	Region served by Calopeia warehouse via cross-fulfilment, with profit of \$265/dr.
Sorange	Yes (85 dr/day)	Yes	B	Sorange	Isolated system; early-game cross-supply from Calopeia covers gap before Sorange factory comes online.
Fardo	Yes (20 dr/day)	Yes	C	Fardo	Entirely self-contained island system.

There were, however, occasions in which both Calopeia and Sorange supported each other: The first was the adoption of a cross-fulfilment policy at the warehouse in Sorange, during the first peak in demand in Calopeia, as at this point demand was significantly lower than capacity in the former, while the latter did not have enough inventory to avoid a stockout; the second occurred in the later stages of the simulation, when the inventory deficit at Sorange had to be covered by the warehouse at Calopeia, whose demand by then would have passed its second peak - The precise timing of this transition was determined algorithmically by the bot, as explained in the subsequent section, which highlights the increased flexibility that resulted from the design and implementation of this tool.

The final network parameter concerned whether warehouses should satisfy demand based on spatial or temporal proximity. The team selected the “nearest” policy across all warehouses for two reasons. First, it ensured that each warehouse served its intended catchment area, preventing Fardo's inventory from being diverted to continental demand (and vice versa). Second, since all continental warehouses maintained large safety stocks under the *Always On* strategy,

there was no meaningful lead-time advantage to selecting 'soonest'; the nearest warehouse was almost always stocked. Cross-fulfilment from the Calopeia warehouse to Entworpe was the only deliberate exception, and this operated correctly under the nearest policy, as no warehouse existed in Entworpe.

Table 9. Mainland network configuration stages

Stage	Period	System	Regions served	Capacity (dr/day)	Total average expected demand (dr/day)
1	730-820	A	- Calopeia - Tyran - Sorange - Entworpe	70	79*
2	820-1200	A	- Calopeia - Tyran - Entworpe	75	71
		B	Sorange	85	63**
3	1201-1460	A	- Calopeia - Tyran - Entworpe - Sorange (from excess stock)	75	81
		B	Sorange	85	110

*Starting stock was 2,448 drums, which made it possible to cover the production deficit in this period.

** Sorange warehouse served Calopeia during its first demand peak

C.2.2. Short-term Decisions

For the network run the parameters over which the bot had control were limited only to reorder point, batch size and priority level. The flowchart showing the decision-making process is included in Appendix C, while a high-resolution version can be found in the supporting documentation.

Re-order Point

In accordance with the *Always On* principle adopted in the overarching strategy, the reorder point (ROP) was set at an artificially high level each day using Equation 10. This calculation ensured that no matter how many goods there were in the pipeline ($I_t + T_t$), production would always be triggered by the simulation software.

However, the bot would also carry out an additional daily verification step. It estimated the total remaining demand and compared it to goods in the warehouse pipeline, through Identity 11. If pipeline inventory exceeded remaining demand, the ROP was set to zero to stop production. In addition, remaining demand was calculated up to day 1450, leaving a 10-day error margin, as it was considered preferable to lose sales in the final 10 days rather than having excess inventory on day 1460.

$$ROP(t) = I_t + T_t + 1000 \quad (10)$$

$$\text{if } I_t + T_t > D_{rem}(t), \text{ then } ROP = 0$$

$$D_{rem}(t) = \sum_{i=t+1}^{1450} \hat{d}^{adj}(i) \quad (11)$$

Batch Size and Production Allocation

The other parameter adjusted daily by the bot was the order size for each warehouse. The objective was to balance between reducing order cost (\$2,000/order) and transport costs against the need to meet demand reliably. While larger orders reduced per-unit logistics costs, they increased the risk that shipments would not be completed and dispatched before warehouse inventory was depleted. This led to the following heuristic rule being implemented during the simulation, based on the results from different trial runs:

$$Q = \begin{cases} 200 & \text{if } T_t \leq 1000 \\ 400 & \text{if } 1000 < T_t \leq 4000 \\ 600 & \text{if } T_t > 4000 \end{cases} \quad (12)$$

As reflected in the batch sizes, the shipment method was always trucks at full capacity. This was feasible by the adopted long-term stock-building strategy, that ensured inventory stability and eliminated the need of just-in-time mail deliveries. This represented another key departure from the strategy adopted in the single region run

The final parameter determined by the bot concerned supply priority between the two warehouses served by the Calopeia factory (System A). By default, all production was allocated to Calopeia, as it had higher demand and was responsible for fulfilling Entworpe's demand. However, if the bot detected that Tyran's inventory had fallen below 400 drums, it increased Tyran's warehouse priority to level 5 (highest) until the safety stock was restored.

This threshold was determined as the optimal point after running multiple simulations that accounted for different levels of demand stochasticity and ensured sufficient safety stock for at least 20 days, based on the estimated average demand.

D. Results and Conclusions

Based on the performance indicators of the single region run, the team was able to validate its overarching strategy: to expand factory capacity slightly above the mean demand, while relying on inventory build-up to avoid stockouts when it peaked. Table 10 shows that this approach led to a 93.87% item fill rate (IFR), with only 77 stockout events in the two-year lifespan of the strategy. It is worth noting that these occurred mostly in the first year, when there had not been enough time to build up stock; for the second peak, which started around day 1146, the high inventory meant that almost all demand was covered. There were only two stockout events after this point, leading to a steep increase in cash reserves. Appendix D contains the graph comparing lost demand and cash balance throughout the run.

However, the lost sales amounted to a total missed profit of approximately \$634,410, which indicated that a slightly larger increase in capacity, of possibly up to 55 dr/day instead of the adopted 50 dr/day, would have led to better overall results. Another finding from this run was that mail deliveries should always be avoided, as this meant a reduction in drum profitability of \$75, almost 25% less than if full trucks were used. Furthermore, the team noticed that the constraint to meeting demand was not the lead time from delivery, but rather factory capacity. This meant that it was better to keep a drum in storage to sell in the second peak for a higher profit, rather than paying a premium for speedy delivery. The low holding costs observed during this run, which amounted to less than 0.5% of the total expenses, also provided further evidence that this approach would be better.

These learnings meant that the strategy adopted for the second run had to be refined in a number of ways. First, capacity would be expanded more aggressively with respect to the overall demand, while still balancing with the capital expense; second, drum profitability should be a priority, which for the network run meant that reducing shipping and fulfilment costs by investing in factories and warehouses could be justified; third, the order quantity should not be fixed at 200 dr/order when inventory levels allowed it, as this would also increase drum profitability; finally, the decision-making algorithm of the bot could be streamlined by only having two alternative modes: build inventory and stop production.

The adoption of this strategy led to a significant improvement in results, as the team not only managed to have the highest final cash among the other groups, but also saw better values across a range of performance indicators. The most important of these is the achieved IFR of 96.88%, with practically almost all stockout events occurring before day 820 – due to infrastructure lead times, it was impossible to reduce this further. It is important to note that this was possible thanks to the significant inventory built in Sorange in the first year of the simulation, which accounts for the extremely low cash balance in the intermediate period of the run, and the operational flexibility provided by the bot. Cross-fulfilment from excess inventory at the warehouse in Calopeia also played a significant role in achieving such a high IFR, underlining the importance of flexibility and adaptability in a logistics network. For a comprehensive view of the cash reserves and lost demand in the network run, refer to Appendix E.

On the other hand, a more efficient end-game strategy could have been pursued, as the warehouse at Fardo had 101 drums left at the end of day 1460, even though at the final stage cross-fulfilment to the mainland was enabled. The cost of this overproduction can be estimated to have been approximately \$98,000. To avoid this problem, the capacity at Fardo could have had a lower value, possibly of 17 dr/day, with the amount invested instead being put into a mainland factory, especially as enabling island to mainland fulfilment reduced profit to \$65/dr, from the optimal \$315/dr.

Overall, the key findings that resulted from the Supply Chain Game can be summarized as follows: when holding costs are low and inventory size is not restricted, production capacity should be at least equal to the mean demand. On many occasions, it can be justified to increase it even further if the demand pattern shows that a surge is expected to occur soon, or the investment lifespan is expected to be short. Second, bulk ordering and transport of goods can lead to a significant reduction in total unit costs; however, these must be balanced with lead times and the specific delivery policies of each case. Third, logistics networks should have a level of resilience that is high enough to absorb demand shocks effectively: One way to achieve this is by allowing individual sub-systems to support one another. Finally, in highly stochastic environments complex forecasting methods have difficulty differentiating signal from noise; this is why simple forecasting methods can deliver superior results. In conclusion, there is no single approach to designing a supply chain, as each problem will have a distinct set of parameters and constraints. Thus, a bespoke approach that capitalizes on these idiosyncrasies is required if one aims to obtain the best results.

Table 10. Simulation performance indicators for days 730-1460.

Performance Indicator	Single region run	Network run
Capital investment	\$1,000,000	\$6,800,000
Total demand	28,743 dr	117,305 dr
Lost demand	2,014 dr	3,655 dr
Item fill rate	93.87%	96.88%
Stockout events	77	30*
Drums sold	26,729 dr	113,650 dr
Leftover inventory	10 dr	101 dr**
Final cash	\$12,375,000	\$25,952,540.52

*Indicator normalized to account for the 5 regions supplied during the network run. The summation of stockout events for all regions is 148.

**All leftover inventory was located in Fardo.

Generative AI statement

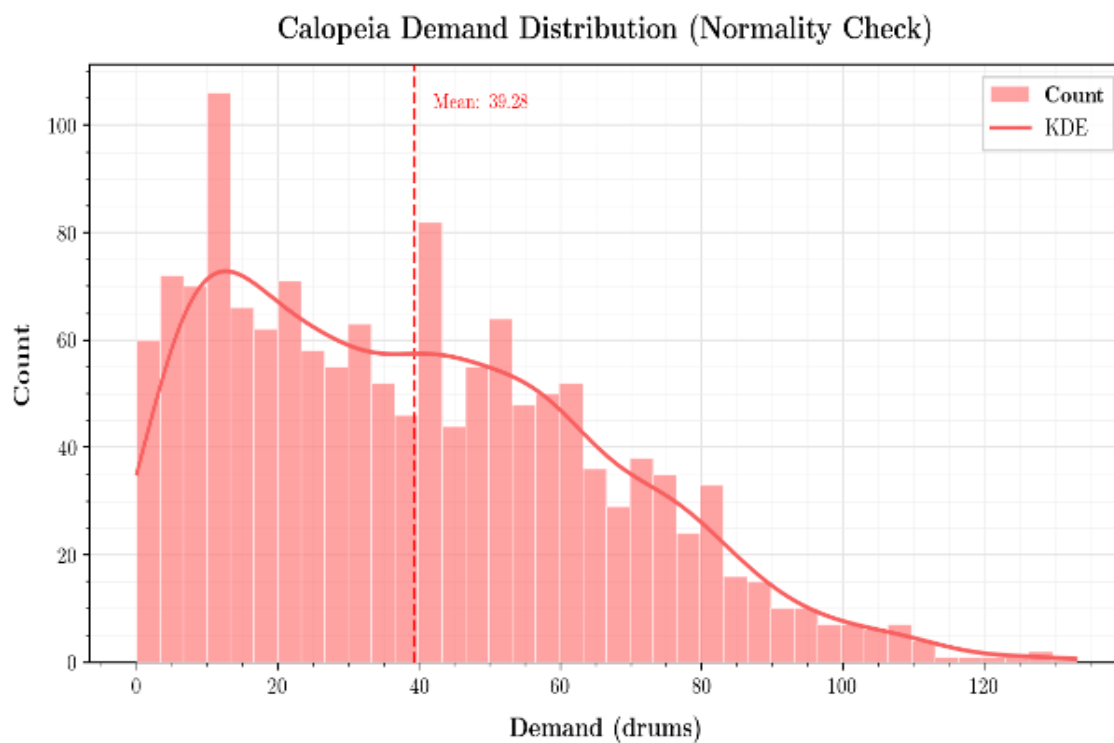
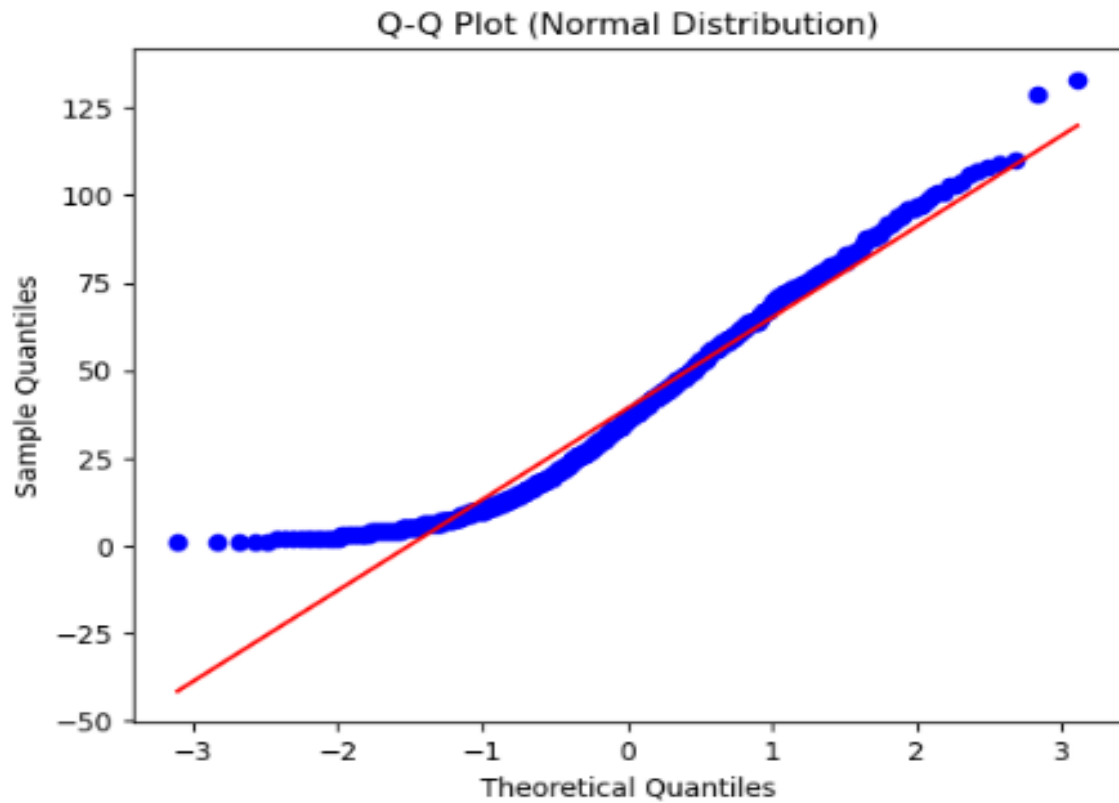
We acknowledge the use of generative AI to aid in specific tasks throughout this project. The Large Language Models (LLM) used were: Gemini Pro 3.0 (Google DeepMind, [Google Gemini](#)) to organize ideas during brainstorming sessions; Claude 4.5 Opus (Anthropic, [Claude](#)), to aid in troubleshooting and the programming of specific algorithms during the setup of the decision-making bot; Claude Sonnet 4.5 (Anthropic, [Claude](#)), to aid in the formatting of graphs; ChatGPT-5 (OpenAI, [ChatGPT](#)) to refine language in the present document, and to organize notes taken during team meetings. We confirm that these tools only played an auxiliary role in the development of this coursework, that all output was validated by the team, and that the content of this report was authored by members of the team.

References

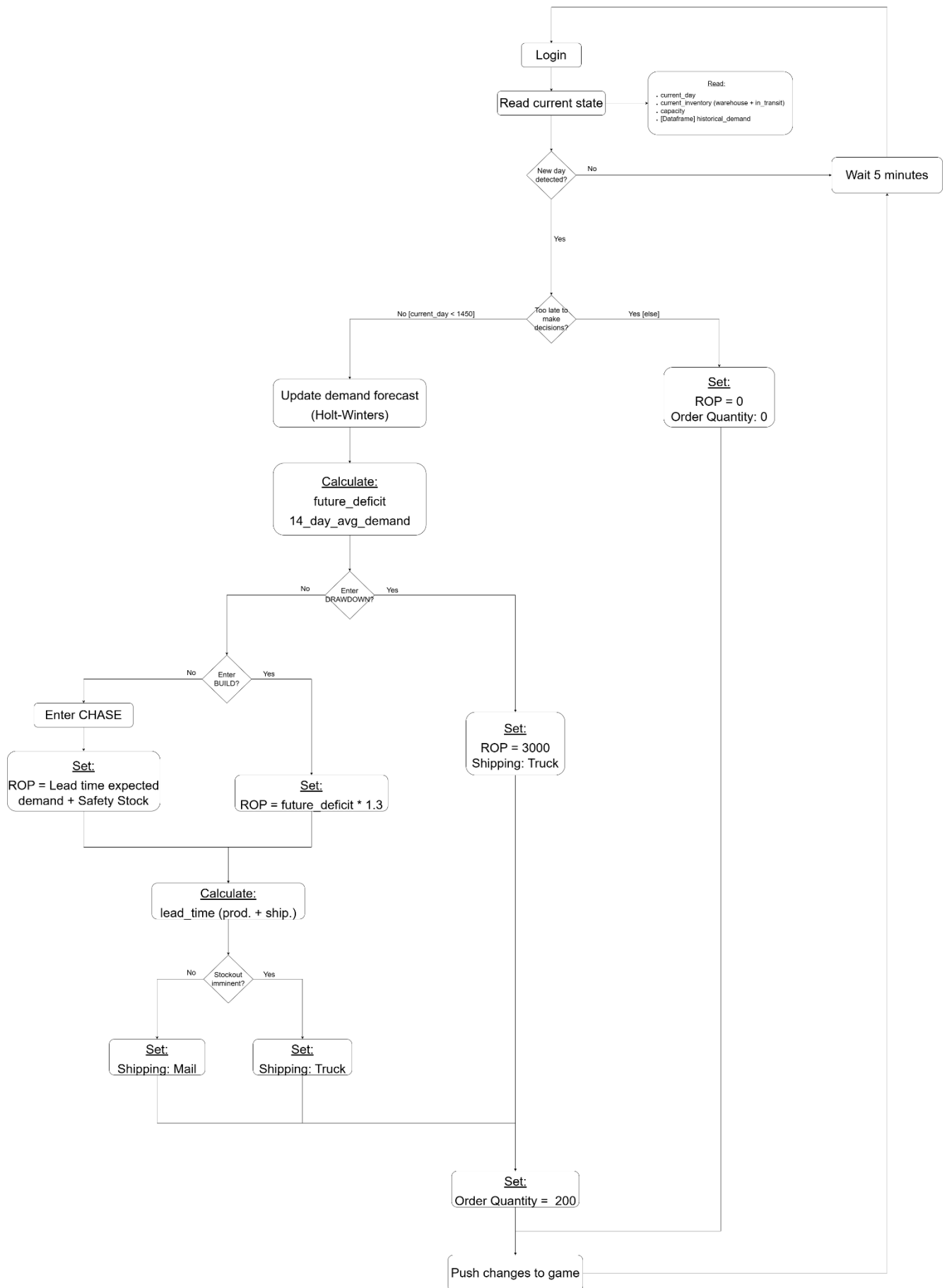
Escribano, J., Angeloudis, P. (2026) *CIVE70108 Freight Transport and Logistics*. Imperial College London. Unpublished teaching material.

Merkle, M., Fusco, A., Jessen, P., Ravi, M. (2022) *Capstone Project, Supply Chain Management Game: Final Report*. Harvard Business School.

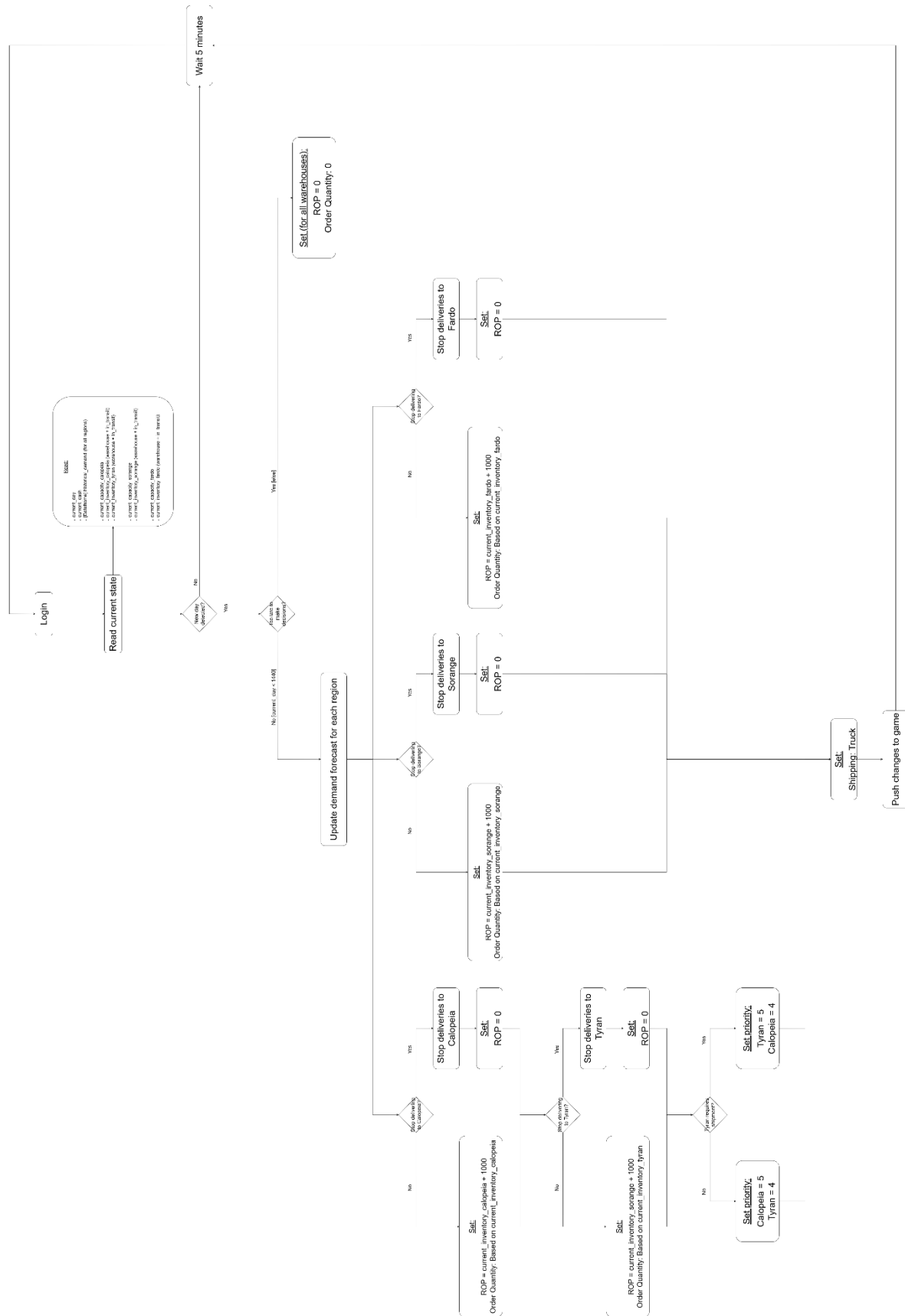
Appendix A: Demand distribution analysis for historical demand in Calopeia



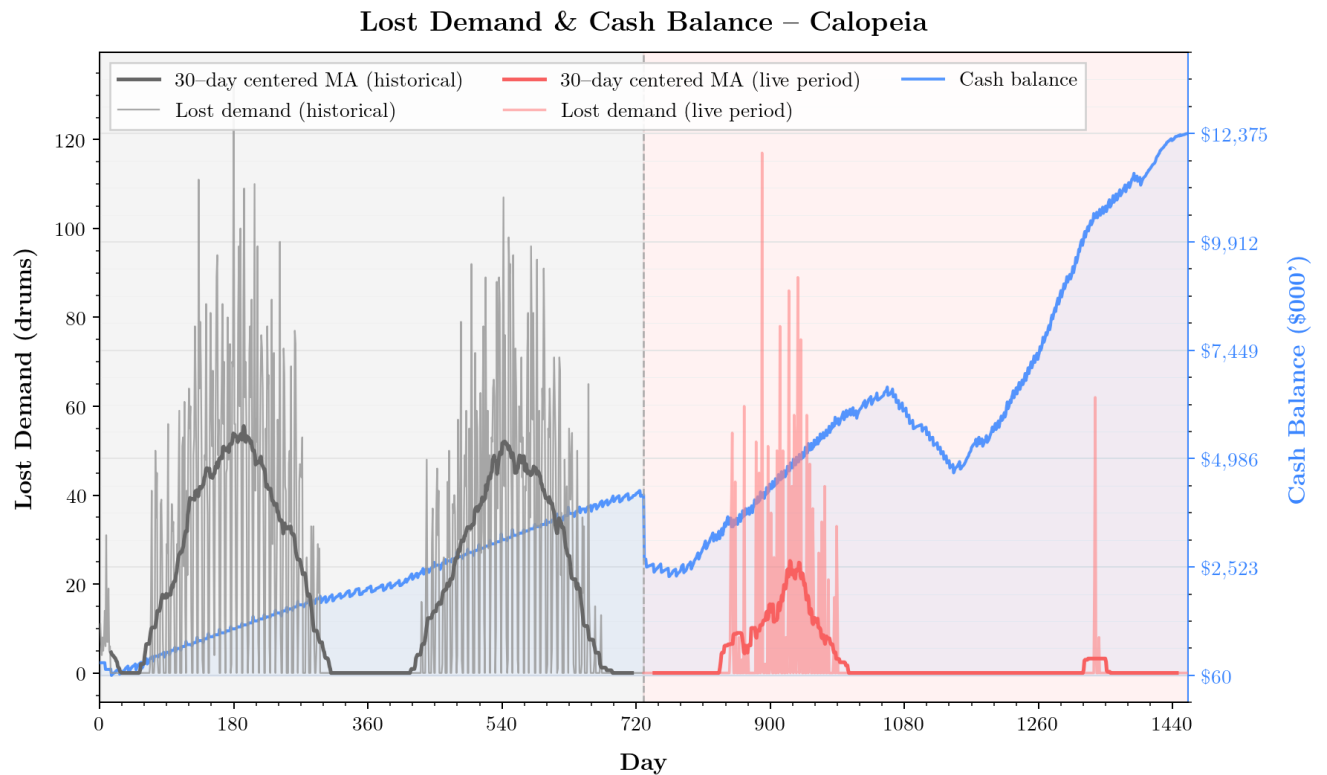
Appendix B: Single region run bot flowchart



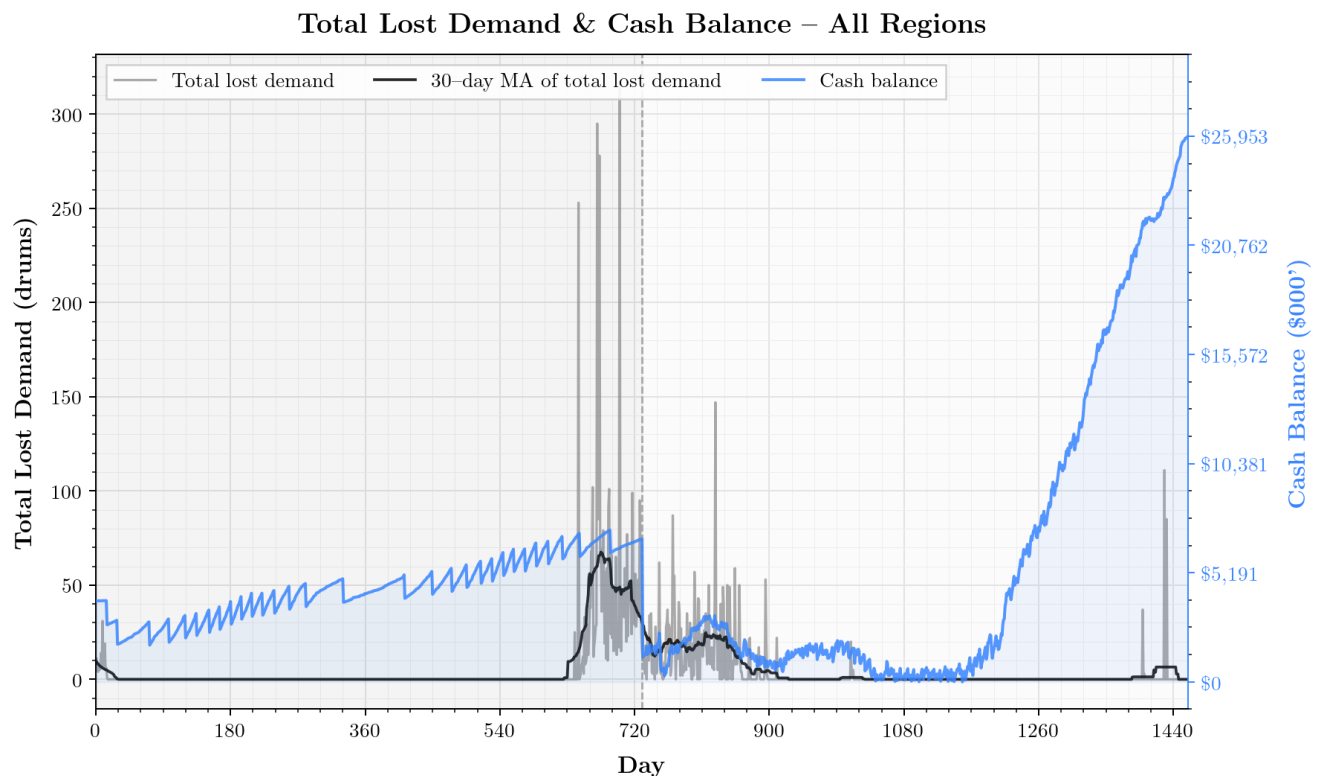
Appendix C: Multi-region network run bot flowchart



Appendix D: Lost demand and cash balance history single region run



Appendix E: Lost demand and cash balance history for network run



Appendix F: Supporting documents and code

The high-resolution version of both flowcharts as well as the bot's code for the single region and network run are provided below. The git repositories of both bots include a quick start guide on how to quickly launch simulations with the respective bots.

Supporting Document	URL
Single region run flowchart	https://drive.google.com/file/d/15azTtSxk8l_D66Mno2EulHsyzcWF9r6A/view?usp=sharing
Network run flowchart	https://drive.google.com/file/d/15azTtSxk8l_D66Mno2EulHsyzcWF9r6A/view?usp=sharing
Single region run git repository	https://github.com/Kilian-S/G4_automation_single_region_run.git
Network run git repository	https://github.com/Kilian-S/G4_automation_network_run.git